an isomorphism for $j \le 2n+1$. As with $\mathbb{R}P^n$, the ring structure in $H^*(L^{2n+1};\mathbb{Z})$ is determined by the ring structure in $H^*(L^{2n+1};\mathbb{Z}_m)$, and likewise for L^∞, where one has the slightly simpler structure $H^*(L^\infty;\mathbb{Z}) \approx \mathbb{Z}[\alpha]/(m\alpha)$ with $|\alpha| = 2$. The case of L^{2n+1} is obtained from this by setting $\alpha^{n+1} = 0$ and adjoining the extra $\mathbb{Z} \approx H^{2n+1}(L^{2n+1};\mathbb{Z})$.

A different derivation of the cup product structure in lens spaces is given in Example 3E.2.

Using the ad hoc notation $H^k_{free}(M)$ for $H^k(M)$ modulo its torsion subgroup, the preceding proposition implies that for a closed orientable manifold M of dimension $2n$, the middle-dimensional cup product pairing $H^n_{free}(M)\times H^n_{free}(M)\to\mathbb{Z}$ is a nonsingular bilinear form on $H^n_{free}(M)$. This form is symmetric or skew-symmetric according to whether n is even or odd. The algebra in the skew-symmetric case is rather simple: With a suitable choice of basis, the matrix of a skew-symmetric nonsingular bilinear form over $\mathbb{Z}$ can be put into the standard form consisting of 2×2 blocks $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ along the diagonal and zeros elsewhere, according to an algebra exercise at the end of the section. In particular, the rank of $H^n(M^{2n})$ must be even when n is odd. We are already familiar with these facts in the case $n = 1$ by the explicit computations of cup products for surfaces in §3.2.

The symmetric case is much more interesting algebraically. There are only finitely many isomorphism classes of symmetric nonsingular bilinear forms over $\mathbb{Z}$ of a fixed rank, but this 'finitely many' grows rather rapidly, for example it is more than 80 million for rank 32; see [Serre 1973] for an exposition of this beautiful chapter of number theory. It is known that for each even $n \ge 2$, every symmetric nonsingular form actually occurs as the cup product pairing in some closed manifold M^{2n}. One can even take M^{2n} to be simply-connected and have the bare minimum of homology: $\mathbb{Z}$'s in dimensions 0 and $2n$ and a $\mathbb{Z}^k$ in dimension n. For $n = 2$ there are at most two nonhomeomorphic simply-connected closed 4-manifolds with the same bilinear form. Namely, there are two manifolds with the same form if the square $\alpha \smile \alpha$ of some $\alpha \in H^2(M^4)$ is an odd multiple of a generator of $H^4(M^4)$, for example for $\mathbb{C}P^2$, and otherwise the M^4 is unique, for example for S^4 or $S^2\times S^2$; see [Freedman & Quinn 1990]. In §4.C we take the first step in this direction by proving a classical result of J. H. C. Whitehead that the homotopy type of a simply-connected closed 4-manifold is uniquely determined by its cup product structure.

Other Forms of Duality

Generalizing the definition of a manifold, an ***n*-manifold with boundary** is a Hausdorff space M in which each point has an open neighborhood homeomorphic either to $\mathbb{R}^n$ or to the half-space $\mathbb{R}^n_+ = \{\,(x_1,\cdots,x_n) \in \mathbb{R}^n \mid x_n \ge 0\,\}$. If a point $x \in M$ corresponds under such a homeomorphism to a point $(x_1,\cdots,x_n) \in \mathbb{R}^n_+$ with $x_n = 0$, then by excision we have $H_n(M, M-\{x\};\mathbb{Z}) \approx H_n(\mathbb{R}^n_+, \mathbb{R}^n_+ - \{0\};\mathbb{Z}) = 0$, whereas if x corresponds to a point $(x_1, \cdots, x_n) \in \mathbb{R}^n_+$ with $x_n > 0$ or to a point of $\mathbb{R}^n$, then $H_n(M, M - \{x\}; \mathbb{Z}) \approx H_n(\mathbb{R}^n, \mathbb{R}^n - \{0\}; \mathbb{Z}) \approx \mathbb{Z}$. Thus the points x with $H_n(M, M - \{x\}; \mathbb{Z}) = 0$ form a well-defined subspace, called the **boundary** of M and denoted ∂M. For example, $\partial \mathbb{R}^n_+ = \mathbb{R}^{n-1}$ and $\partial D^n = S^{n-1}$. It is evident that ∂M is an $(n-1)$-dimensional manifold with empty boundary.

If M is a manifold with boundary, then a **collar** neighborhood of ∂M in M is an open neighborhood homeomorphic to $\partial M \times [0,1)$ by a homeomorphism taking ∂M to $\partial M \times \{0\}$.

Proposition 3.42. *If M is a compact manifold with boundary, then ∂M has a collar neighborhood.*

Proof: Let M' be M with an external collar attached, the quotient of the disjoint union of M and $\partial M \times [0,1]$ in which $x \in \partial M$ is identified with $(x, 0) \in \partial M \times [0,1]$. It will suffice to construct a homeomorphism $h : M \to M'$ since $\partial M'$ clearly has a collar neighborhood.

Since M is compact, so is the closed subspace ∂M. This implies that we can choose a finite number of continuous functions $\varphi_i : \partial M \to [0,1]$ such that the sets $V_i = \varphi_i^{-1}(0,1]$ form an open cover of ∂M and each V_i has closure contained in an open set $U_i \subset M$ homeomorphic to the half-space $\mathbb{R}^n_+$. After dividing each φ_i by $\sum_j \varphi_j$ we may assume $\sum_i \varphi_i = 1$.

Let $\psi_k = \varphi_1 + \cdots + \varphi_k$ and let $M_k \subset M'$ be the union of M with the points $(x,t) \in \partial M \times [0,1]$ with $t \le \psi_k(x)$. By definition $\psi_0 = 0$ and $M_0 = M$. We construct a homeomorphism $h_k : M_{k-1} \to M_k$ as follows. The homeomorphism $U_k \approx \mathbb{R}^n_+$ gives a collar neighborhood $\partial U_k \times [-1, 0]$ of ∂U_k in U_k, with $x \in \partial U_k$ corresponding to $(x, 0) \in \partial U_k \times [-1, 0]$. Via the external collar $\partial M \times [0,1]$ we then have an embedding $\partial U_k \times [-1, 1] \subset M'$. We define h_k to be the identity outside this $\partial U_k \times [-1,1]$, and for $x \in \partial U_k$ we let h_k stretch the segment $\{x\} \times [-1, \psi_{k-1}(x)]$ linearly onto $\{x\} \times [-1, \psi_k(x)]$. The composition of all the h_k's then gives a homeomorphism $M \approx M'$, finishing the proof. □

More generally, collars can be constructed for the boundaries of paracompact manifolds in the same way.

A compact manifold M with boundary is defined to be R-orientable if $M - \partial M$ is R-orientable as a manifold without boundary. If $\partial M \times [0,1)$ is a collar neighborhood of ∂M in M then $H_i(M, \partial M; R)$ is naturally isomorphic to $H_i(M - \partial M, \partial M \times (0, \varepsilon); R)$, so when M is R-orientable, Lemma 3.27 gives a relative fundamental class $[M]$ in $H_n(M, \partial M; R)$ restricting to a given orientation at each point of $M - \partial M$.

It will not be difficult to deduce the following generalization of Poincaré duality to manifolds with boundary from the version we have already proved for noncompact manifolds:

Theorem 3.43. *Suppose M is a compact R-orientable n-manifold whose boundary ∂M is decomposed as the union of two compact $(n-1)$-dimensional manifolds A and B with a common boundary $\partial A = \partial B = A\cap B$. Then cap product with a fundamental class $[M] \in H_n(M,\partial M;R)$ gives isomorphisms $D_M : H^k(M,A;R) \to H_{n-k}(M,B;R)$ for all k.*

The possibility that A, B, or $A\cap B$ is empty is not excluded. The cases $A = \varnothing$ and $B = \varnothing$ are sometimes called Lefschetz duality.

Proof: The cap product map $D_M : H^k(M,A;R) \to H_{n-k}(M,B;R)$ is defined since the existence of collar neighborhoods of $A\cap B$ in A and B and ∂M in M implies that A and B are deformation retracts of open neighborhoods U and V in M such that $U\cup V$ deformation retracts onto $A\cup B = \partial M$ and $U\cap V$ deformation retracts onto $A\cap B$.

The case $B = \varnothing$ is proved by applying Theorem 3.35 to $M - \partial M$. Via a collar neighborhood of ∂M we see that $H^k(M,\partial M;R) \approx H^k_c(M-\partial M;R)$, and there are obvious isomorphisms $H_{n-k}(M;R) \approx H_{n-k}(M-\partial M;R)$.

The general case reduces to the case $B = \varnothing$ by applying the five-lemma to the following diagram, where coefficients in R are implicit:

$$\begin{tikzcd}
\cdots \arrow[r] & H^k(M,\partial M) \arrow[r] \arrow[dd, "{[M]\frown}"] & H^k(M,A) \arrow[r] \arrow[dd, "{[M]\frown}"] & H^k(\partial M,A) \arrow[r] \arrow[d, "\approx"] & H^{k+1}(M,\partial M) \arrow[r] \arrow[dd, "{[M]\frown}"] & \cdots \\
 & & & H^k(B,\partial B) \arrow[d, "{[B]\frown}"] & & \\
\cdots \arrow[r] & H_{n-k}(M) \arrow[r] & H_{n-k}(M,B) \arrow[r] & H_{n-k-1}(B) \arrow[r] & H_{n-k-1}(M) \arrow[r] & \cdots
\end{tikzcd}$$

For commutativity of the middle square one needs to check that the boundary map $H_n(M,\partial M) \to H_{n-1}(\partial M)$ sends a fundamental class for M to a fundamental class for ∂M. We leave this as an exercise at the end of the section. $\square$

Next we turn to **Alexander duality**:

Theorem 3.44. *If K is a compact, locally contractible, nonempty, proper subspace of S^n, then $\widetilde{H}_i(S^n - K;\mathbb{Z}) \approx \widetilde{H}^{n-i-1}(K;\mathbb{Z})$ for all i.*

The special case that K is a sphere or disk was treated by more elementary means in Proposition 2B.1. As remarked there, it is interesting that the homology of $S^n - K$ does not depend on the way that K is embedded in S^n. There can be local pathologies as in the case of the Alexander horned sphere, or global complications as with knotted circles in S^3, but these have no effect on the homology of the complement. The only requirement is that K is not too bad a space itself. An example where the theorem fails without the local contractibility assumption is the 'quasi-circle,' defined in an exercise for §1.3. This compact subspace $K \subset \mathbb{R}^2$ can be regarded as a subspace of

S^2 by adding a point at infinity. Then we have $\widetilde{H}_0(S^2-K;\mathbb{Z})\approx\mathbb{Z}$ since S^2-K has two path-components, but $\widetilde{H}^1(K;\mathbb{Z})=0$ since K is simply-connected.

Proof: We will obtain the desired isomorphism when $i\neq 0$ as the composition of five isomorphisms

$$\begin{aligned} H_i(S^n-K) &\approx H_c^{n-i}(S^n-K) \\ &\approx \varinjlim H^{n-i}(S^n-K,U-K) \\ &\approx \varinjlim H^{n-i}(S^n,U) \\ &\approx \varinjlim \widetilde{H}^{n-i-1}(U) \quad \text{if } i\neq 0 \\ &\approx \widetilde{H}^{n-i-1}(K) \end{aligned}$$

where coefficients in $\mathbb{Z}$ will be implicit throughout the proof, and the direct limits are taken with respect to open neighborhoods U of K. The first isomorphism is Poincaré duality. The second is the definition of cohomology with compact supports. The third is excision. The fourth comes from the long exact sequences of the pairs (S^n,U). For the final isomorphism, an easy special case is when K has a neighborhood that is a mapping cylinder of some map $X\to K$, as in the 'letter examples' at the beginning of Chapter 0, since in this case we can compute the direct limit using neighborhoods U which are segments of the mapping cylinder that deformation retract to K.

To obtain the last isomorphism in the general case we need to quote Theorem A.7 in the Appendix, which says that K is a retract of some neighborhood U_0 in S^n since K is locally contractible. In computing the direct limits we can then restrict attention to open sets $U\subset U_0$, which all retract to K by restricting the retraction of U_0. This implies that the natural restriction map $\varinjlim H^*(U)\to H^*(K)$ is surjective since we can pull back each element of $H^*(K)$ to the direct limit via the retractions $U\to K$.

To see injectivity of the map $\varinjlim H^*(U)\to H^*(K)$, we first show that each neighborhood $U\subset U_0$ of K contains a neighborhood V such that the inclusion $V\hookrightarrow U$ is homotopic to the retraction $V\to K\subset U$. Namely, regarding U as a subspace of an $\mathbb{R}^n\subset S^n$, the linear homotopy $U\times I\to\mathbb{R}^n$ from the identity to the retraction $U\to K$ takes $K\times I$ to K, hence takes $V\times I$ to U for some neighborhood V of K, by compactness of I. Since the inclusion $V\hookrightarrow U$ is homotopic to the retraction $V\to K\subset U$, the restriction $H^*(U)\to H^*(V)$ factors through $H^*(K)$, and therefore if an element of $H^*(U)$ restricts to zero in $H^*(K)$, it restricts to zero in $H^*(V)$. This implies that the map $\varinjlim H^*(U)\to H^*(K)$ is injective.

The only difficulty in the case $i=0$ is that the fourth of the five isomorphisms above does not hold, and instead we have only a short exact sequence

$$0\longrightarrow \widetilde{H}^{n-i-1}(U)\longrightarrow H^{n-i}(S^n,U)\longrightarrow \widetilde{H}^{n-i}(S^n)\longrightarrow 0$$

To get around this little problem, observe that all the groups involved in the first three of the five isomorphisms map naturally to the corresponding groups with K and U empty. Then if we take the kernels of these maps we get an isomorphism

$\widetilde{H}^0(S^n - K) \approx \varinjlim \widetilde{H}^n(U)$, and we have seen that the latter group is isomorphic to $\widetilde{H}^n(K)$. □

Corollary 3.45. *If $X \subset \mathbb{R}^n$ is compact and locally contractible then $H_i(X;\mathbb{Z})$ is 0 for $i \geq n$ and torsionfree for $i = n-1$ and $n-2$.*

For example, a closed nonorientable n-manifold M cannot be embedded as a subspace of $\mathbb{R}^{n+1}$ since $H_{n-1}(M;\mathbb{Z})$ contains a $\mathbb{Z}_2$ subgroup, by Corollary 3.28. Thus the Klein bottle cannot be embedded in $\mathbb{R}^3$. More generally, the 2-dimensional complex $X_{m,n}$ studied in Example 1.24, the quotient spaces of $S^1 \times I$ under the identifications $(z,0) \sim (e^{2\pi i/m}z, 0)$ and $(z,1) \sim (e^{2\pi i/n}z, 1)$, cannot be embedded in $\mathbb{R}^3$ if m and n are not relatively prime, since $H_1(X_{m,n}\mathbb{Z})$ is $\mathbb{Z}\times\mathbb{Z}_d$ where d is the greatest common divisor of m and n. The Klein bottle is the case $m = n = 2$.

Proof: Viewing X as a subspace of the one-point compactification S^n, Alexander duality gives isomorphisms $\widetilde{H}^i(X;\mathbb{Z}) \approx \widetilde{H}_{n-i-1}(S^n - X;\mathbb{Z})$. The latter group is zero for $i \geq n$ and torsionfree for $i = n-1$, so the result follows from the universal coefficient theorem since X has finitely generated homology groups. □

Here is another kind of duality which generalizes the calculation of the local homology groups $H_i(\mathbb{R}^n, \mathbb{R}^n - \{x\};\mathbb{Z})$:

Proposition 3.46. *If K is a compact, locally contractible subspace of an orientable n-manifold M, then there are isomorphisms $H_i(M, M-K;\mathbb{Z}) \approx H^{n-i}(K;\mathbb{Z})$ for all i.*

Proof: Let U be an open neighborhood of K in M and let V be the complement of a compact set in M. We assume $U \cap V = \varnothing$. Then cap product with fundamental classes gives a commutative diagram with exact rows

$$\begin{tikzcd}
\cdots \arrow[r] & H_i(M-K) \arrow[r] & H_i(M) \arrow[r] & H_i(M,M-K) \approx H_i(U,U-K) \arrow[r] & \cdots \\
\cdots \arrow[r] & H^{n-i}(M,U\cup V) \arrow[r] \arrow[u] & H^{n-i}(M,V) \arrow[r] \arrow[u] & H^{n-i}(U\cup V,V) \approx H^{n-i}(U) \arrow[r] \arrow[u] & \cdots
\end{tikzcd}$$

Passing to the direct limit over decreasing $U \supset K$ and V, the first two vertical arrows become the Poincaré duality isomorphisms $H_i(M-K) \approx H_c^{n-i}(M-K)$ and $H_i(M) \approx H_c^{n-i}(M)$. The five-lemma then gives an isomorphism $H_i(M, M-K) \approx \varinjlim H^{n-i}(U)$. The latter group will be isomorphic to $H^{n-i}(K)$ by the argument in the proof of Theorem 3.44, provided that K is a retract of some neighborhood in M. To obtain such a retraction we can first construct a map $M \hookrightarrow \mathbb{R}^k$ that is an embedding near the compact set K, for some large k, by the method of Corollary A.9 in the Appendix. Then a neighborhood of K in $\mathbb{R}^k$ retracts onto K by Theorem A.7 in the Appendix, so the restriction of this retraction to a neighborhood of K in M finishes the job. □

There is a way of extending Alexander duality and the duality in the preceding proposition to compact sets K that are not locally contractible, by replacing the singular cohomology of K with another kind of cohomology called Čech cohomology. This is defined in the following way. To each open cover $\mathcal{U} = \{U_\alpha\}$ of a given space X we can associate a simplicial complex $N(\mathcal{U})$ called the **nerve** of $\mathcal{U}$. This has a vertex v_α for each U_α, and a set of $k+1$ vertices spans a k-simplex whenever the $k+1$ corresponding U_α's have nonempty intersection. When another cover $\mathcal{V} = \{V_\beta\}$ is a refinement of $\mathcal{U}$, so each V_β is contained in some U_α, then these inclusions induce a simplicial map $N(\mathcal{V}) \to N(\mathcal{U})$ that is well-defined up to homotopy. We can then form the direct limit $\varinjlim H^i(N(\mathcal{U});G)$ with respect to finer and finer open covers $\mathcal{U}$. This limit group is by definition the **Čech cohomology group** $\check{H}^i(X;G)$. For a full exposition of this cohomology theory see [Eilenberg & Steenrod 1952]. With an analogous definition of relative groups, Čech cohomology turns out to satisfy the same axioms as singular cohomology, and indeed a stronger form of excision: a map $(X,A) \to (Y,B)$ that restricts to a homeomorphism $X - A \to Y - B$ induces isomorphisms on Čech cohomology groups. For spaces homotopy equivalent to CW complexes, Čech cohomology coincides with singular cohomology, but for spaces with local complexities it often behaves more reasonably. For example, if X is the subspace of $\mathbb{R}^3$ consisting of the spheres of radius $1/n$ and center $(1/n, 0, 0)$ for $n = 1, 2, \cdots$, then contrary to what one might expect, $H^3(X;\mathbb{Z})$ is nonzero, as shown in [Barratt & Milnor 1962]. But $\check{H}^3(X;\mathbb{Z}) = 0$ and $\check{H}^2(X;\mathbb{Z}) = \mathbb{Z}^\infty$, the direct sum of countably many copies of $\mathbb{Z}$.

Oddly enough, the corresponding Čech homology groups defined using inverse limits are not so well-behaved. This is because the exactness axiom fails due to the algebraic fact that an inverse limit of exact sequences need not be exact, as a direct limit would be; see §3.F. However, there is a way around this problem using a more refined definition. This is Steenrod homology theory, which the reader can find out about in [Milnor 1995].

Exercises

1. Show that there exist nonorientable 1-dimensional manifolds if the Hausdorff condition is dropped from the definition of a manifold.

2. Show that deleting a point from a manifold of dimension greater than 1 does not affect orientability of the manifold.

3. Show that every covering space of an orientable manifold is an orientable manifold.

4. Given a covering space action of a group G on an orientable manifold M by orientation-preserving homeomorphisms, show that M/G is also orientable.

5. Show that $M \times N$ is orientable iff M and N are both orientable.

6. Given two disjoint connected n-manifolds M_1 and M_2, a connected n-manifold $M_1 \sharp M_2$, their *connected sum*, can be constructed by deleting the interiors of closed n-balls $B_1 \subset M_1$ and $B_2 \subset M_2$ and identifying the resulting boundary spheres ∂B_1 and ∂B_2 via some homeomorphism between them. (Assume that each B_i embeds nicely in a larger ball in M_i.)